The Dot Product

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Computing dot products and angles**

Let $\mathbf{u} = \langle 2, 5\rangle$ and $\mathbf{v} = \langle -3, 4\rangle$. Find:

a $\mathbf{u} \cdot \mathbf{v}$ $(2)(-3) + (5)(4) = 14$.

b $|\mathbf{u}|$ and $|\mathbf{v}|$ $|\mathbf{u}| = \sqrt{29}$, $|\mathbf{v}| = 5$.

c the angle between $\mathbf{u}$ and $\mathbf{v}$, to one decimal place $\cos\theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7°$.

(a) $(2)(-3) + (5)(4) = 14$.; (b) $|\mathbf{u}| = \sqrt{29}$, $|\mathbf{v}| = 5$.; (c) $\cos\theta = \frac{14}{5\sqrt{29}} \approx 0.5199$, so $\theta \approx 58.7°$.

2 **[Foundational]** • **Target: Computing dot products and angles**

Determine whether each pair of vectors is orthogonal:

a $\langle 4, -6\rangle$ and $\langle 3, 2\rangle$ $12 - 12 = 0$: orthogonal. **b** $\langle 1, 2\rangle$ and $\langle 2, 1\rangle$ $2 + 2 = 4 \neq 0$: not orthogonal.

(a) $12 - 12 = 0$: orthogonal.; (b) $2 + 2 = 4 \neq 0$: not orthogonal.

3 **[Foundational]** • **Target: Computing dot products and angles**

Find the value of k so that $\langle k, 3\rangle$ is orthogonal to $\langle 2, -4\rangle$.

$2k - 12 = 0$, so $k = 6$.

4 **[Foundational]** • **Target: Projections and work**

A force $\mathbf{F} = \langle 8, 5\rangle$ N moves an object along the displacement $\mathbf{d} = \langle 10, 0\rangle$ m. Find the work done.

$W = \mathbf{F} \cdot \mathbf{d} = 80$ J.

5 **[Foundational]** • **Target: Projections and work**

A wagon is pulled a distance of 20 m by a rope making an angle of $30°$ with the ground, with a force of magnitude 50 N. Find the work done, to one decimal place.

$W = |\mathbf{F}||\mathbf{d}|\cos\theta = 50(20)\cos 30° \approx 1000(0.8660) \approx 866.0$ J.

6 **[Foundational]** • **Target: Properties of the dot product**

Let $\mathbf{u} = \langle 3, 1\rangle$, $\mathbf{v} = \langle -2, 4\rangle$, $\mathbf{w} = \langle 1, 1\rangle$. Verify that $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$ (the distributive property).

$\mathbf{v} + \mathbf{w} = \langle -1, 5\rangle$: $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = -3 + 5 = 2$. Separately: $\mathbf{u} \cdot \mathbf{v} = -6 + 4 = -2$, $\mathbf{u} \cdot \mathbf{w} = 3 + 1 = 4$; sum $= 2$. Both equal 2. ✓

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7 [Foundational] • Target: Angle classification

Without finding the angle exactly, classify each pair of vectors as forming an acute, right, or obtuse angle, using only the sign of the dot product:

a $\langle 3, 5\rangle$ and $\langle 2, -1\rangle$ Dot product $= 6 - 5 = 1 > 0$: acute.

b $\langle -1, 4\rangle$ and $\langle 8, 2\rangle$ Dot product $= -8 + 8 = 0$: right angle.

c $\langle 2, 2\rangle$ and $\langle -3, -3\rangle$ Dot product $= -6 - 6 = -12 < 0$: obtuse.

(a) Dot product $= 6 - 5 = 1 > 0$: acute.; (b) Dot product $= -8 + 8 = 0$: right angle.; (c) Dot product $= -6 - 6 = -12 < 0$: obtuse.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Scalar projection and further applications

Find the scalar projection of $\mathbf{u} = \langle 5, -2\rangle$ onto $\mathbf{v} = \langle 3, 4\rangle$ (the signed length $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{\mathbf{u}\cdot\mathbf{v}}{|\mathbf{v}|}$), to two decimal places.

$\mathbf{u}\cdot\mathbf{v} = 15 - 8 = 7$. $|\mathbf{v}| = 5$. $\text{comp}_{\mathbf{v}}\mathbf{u} = \frac{7}{5} = 1.40$.

9 [Intermediate] • Target: Scalar projection and further applications

A sled is pulled by a rope at $40°$ above horizontal with a force of 120 N, moving the sled 15 m horizontally. Find the work done by the horizontal component of the force.

$W = |\mathbf{F}|\cos(40°)\cdot d = 120\cos 40°\cdot 15 \approx 120(0.766)(15) \approx 1378.8$ J.

10 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Decompose $\mathbf{u} = \langle 7, 4\rangle$ into a component parallel to $\mathbf{v} = \langle 1, 0\rangle$ and a component perpendicular to $\mathbf{v}$.

Parallel: $\text{proj}_{\mathbf{v}}\mathbf{u} = \frac{7}{1}\langle 1, 0\rangle = \langle 7, 0\rangle$. Perpendicular: $\mathbf{u} - \langle 7, 0\rangle = \langle 0, 4\rangle$. Check: $\langle 7, 0\rangle + \langle 0, 4\rangle = \langle 7, 4\rangle$ ✓, and $\langle 0, 4\rangle \cdot \langle 1, 0\rangle = 0$ ✓.

11 [Intermediate] • Target: Deriving the dot product angle formula

State the formula relating the dot product to the angle between two vectors, and explain why $\cos\theta = 0$ corresponds to orthogonal vectors.

$\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$. Since $|\mathbf{u}|, |\mathbf{v}| > 0$ for nonzero vectors, the dot product is zero exactly when $\cos\theta = 0$, i.e. $\theta = 90°$ — perpendicular vectors.

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12 [Intermediate] • Target: The Cauchy-Schwarz inequality (preview)

Explain why $|\mathbf{u}\cdot\mathbf{v}| \le |\mathbf{u}||\mathbf{v}|$ must always hold, using the formula $\mathbf{u}\cdot\mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos\theta$.

Since $-1 \le \cos\theta \le 1$ for any angle, multiplying by the non-negative quantity $|\mathbf{u}||\mathbf{v}|$ gives $-|\mathbf{u}||\mathbf{v}| \le \mathbf{u}\cdot\mathbf{v} \le |\mathbf{u}||\mathbf{v}|$, which is exactly $|\mathbf{u}\cdot\mathbf{v}| \le |\mathbf{u}||\mathbf{v}|$.

13 [Intermediate] • Target: Dot products with variable components

Find all values of k such that $\langle k, 3\rangle$ and $\langle k, -3\rangle$ are orthogonal.

Dot product: $k^2 - 9 = 0 \Rightarrow k = \pm 3$.

14 [Intermediate] • Target: Dot products with variable components

Find all values of k such that the angle between $\langle 1, k\rangle$ and $\langle 2, 1\rangle$ is $45°$.

$\cos 45° = \frac{2+k}{\sqrt{1+k^2}\sqrt{5}} = \frac{\sqrt{2}}{2}$. Squaring: $\frac{(2+k)^2}{5(1+k^2)} = \frac{1}{2}$, so $2(2+k)^2 = 5(1+k^2)$: $2(4+4k+k^2) = 5+5k^2$, $8+8k+2k^2 = 5+5k^2$, $3k^2 - 8k - 3 = 0$. Quadratic formula: $k = \frac{8 \pm \sqrt{64+36}}{6} = \frac{8 \pm 10}{6}$: $k = 3$ or $k = -\frac{1}{3}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 1.02 Domain and Range

A weather vane experiences wind force $\mathbf{F} = \langle 15, -8\rangle$ N. The vane can only rotate about a pivot, so only the component of force perpendicular to the vane's current direction $\mathbf{d} = \langle 3, 4\rangle$ causes it to turn. Find the magnitude of the perpendicular (turning) component of the force, using the fact that the perpendicular component's magnitude equals $\frac{|\mathbf{F}\times\mathbf{d}|}{|\mathbf{d}|}$ in 2D (using the scalar cross product $F_x d_y - F_y d_x$), and separately confirm the parallel component using the dot product.

Scalar cross product: $F_x d_y - F_y d_x = 15(4) - (-8)(3) = 60 + 24 = 84$. Perpendicular component magnitude $= \frac{84}{5} = 16.8$ N. Parallel component via dot product: $\mathbf{F}\cdot\mathbf{d} = 45 - 32 = 13$; magnitude $= \frac{13}{5} = 2.6$ N. Together these decompose the total force into turning and non-turning parts.

16 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 1.03 Average Rate of Change

Two salespeople's monthly sales across three product categories are given by vectors $\mathbf{s}_1 = \langle 40, 25, 10\rangle$ and $\mathbf{s}_2 = \langle 15, 30, 45\rangle$ (units sold). Find $\mathbf{s}_1 \cdot \mathbf{s}_2$ and explain what this dot product does NOT directly tell you about their relative performance (versus what the individual sums would show).

$\mathbf{s}_1 \cdot \mathbf{s}_2 = 40(15) + 25(30) + 10(45) = 600 + 750 + 450 = 1800$. This dot product is a single combined measure of alignment/overlap in their sales patterns, but it does NOT directly reveal who sold more in total (that would come from summing each vector's own components: 75 vs 90 units) — the dot product measures a different kind of relationship (similarity in distribution across categories), not total volume.

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17 [Challenging] • Target: Deriving the dot product angle formula • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Level 2 · MediumVector $\vec{u} = \langle 4, -3 \rangle$ and vector $\vec{v} = \langle 1, 7 \rangle$. Calculate the magnitude of each vector, determine the dot product $\vec{u} \cdot \vec{v}$, and find the exact angle θ between them in radians.

Magnitudes: $|\vec{u}| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. $|\vec{v}| = \sqrt{1^2 + 7^2} = \sqrt{1 + 49} = \sqrt{50} = 5\sqrt{2}$. Dot Product: $\vec{u} \cdot \vec{v} = (4)(1) + (-3)(7) = 4 - 21 = -17$. Angle Between Vectors: Use $\cos\theta = \frac{\vec{u} \cdot \vec{v}}{|\vec{u}||\vec{v}|}$: $\cos\theta = \frac{-17}{(5)(5\sqrt{2})} = \frac{-17}{25\sqrt{2}} = \frac{-17\sqrt{2}}{50} \approx -0.48083..$

18 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the exact angle θ between $\mathbf{u} = \langle 3, 1 \rangle$ and $\mathbf{v} = \langle 1, 2 \rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.

Step 1: Compute dot product and magnitudes: $\mathbf{u} \cdot \mathbf{v} = 3(1) + 1(2) = 5$, $||\mathbf{u}|| = \sqrt{10}$, $||\mathbf{v}|| = \sqrt{5}$. Step 2: Angle between vectors: $\cos\theta = \frac{5}{\sqrt{10}\sqrt{5}} = \frac{5}{5\sqrt{2}} = \frac{\sqrt{2}}{2} \Rightarrow \boldsymbol{\theta = \frac{\pi}{4} = 45^\circ}$. Step 3: Vector projection onto $\mathbf{v}$.

19 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

[Trigonometry & Triangle Laws · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Computing dot products and angles

Lines $\ell_1 \parallel \ell_2$ cut by a transversal form corresponding angles of $(4x - 3)^\circ$ and $(3x + 4)^\circ$. Find the measure of the angle $(4x - 3)^\circ$.

a. 26

b. 34

c. **25** ✓

d. 20

$4x - 3 = 3x + 4 \Rightarrow x = 7$. Angle $= 4(7) - 3 = 25^\circ$.

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22 [Foundational] • Target: Projections and work

In ΔABC, the side lengths opposite to angles A, B, and C are a, b, and c, respectively. If $\angle A - \angle B = 90^\circ$ and $a + b = \sqrt{2}c$, what is the measure of $\angle C$?

a. 30°

b. 45°

c. 60° ✓

d. 75°

We synthesize the Law of Sines with trigonometric sum-to-product and double-angle identities. 1. By the Law of Sines, $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$, which allows us to rewrite the given side relation in terms of angles: $\frac{a+b}{c} = \frac{\sin A + \sin B}{\sin C} = \sqrt{2}$ 2. Applying the sum-to-product identity to the numerator: $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$ Since the angles of a triangle sum to 180°.

23 [Foundational] • Target: Properties of the dot product

In the diagram below, line AB is parallel to line CD. Transversal line EF intersects AB at point G and CD at point H. If the measure of angle AGH is 50°, what is the measure of angle GHD?

a. 50°

b. **130°** ✓

c. 40°

d. 90°

When a transversal intersects two parallel lines, consecutive interior angles are supplementary, meaning they add up to 180 degrees. Angle AGH and angle GHC are consecutive interior angles. Therefore, $m\angle AGH + m\angle GHC = 180^\circ$. Since $m\angle AGH = 50^\circ$, we have $50^\circ + m\angle GHC = 180^\circ$. Solving for $m\angle GHC$ gives $m\angle GHC = 180^\circ - 50^\circ = 130^\circ$. Angle GHD and angle GHC are angles on a straight line, forming a linear pair. Thus.

24 [Foundational] • Target: Angle classification

In the figure, right triangle ABC has a right angle at vertex C, and altitude CD is drawn perpendicular to hypotenuse AB. If $AD = x$, $DB = x + 5$, and $CD = 6$, what is the length of hypotenuse AB?

a. 4

b. 12

c. **13** ✓

d. 23

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ Substitute the given values into the equation: $6^2 = x(x+5)$ $36 = x^2 + 5x$ $x^2 + 5x - 36 = 0$ Factor the quadratic equation: $(x+9)(x-4) = 0$ This yields $x = -9$ or $x = 4$. Because x represents a geometric length, it must be positive, so $x = 4$. The length of hypotenuse AB is the sum of segments AD and DB.

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25 [Foundational] • Target: Scalar projection and further applications

A particle of mass 5 kg moves under the action of a variable force $\mathbf{F}$ N. The force is given by $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ N, where x is the displacement in metres from a fixed origin O. The particle moves from $x = 2$ m to $x = 4$ m. Find the work done by the force $\mathbf{F}$ during this displacement.

a. 32 J

b. 44 J

c. 56 J ✓

d. 68 J

The work done by a variable force $\mathbf{F}$ when a particle moves from position x_1 to x_2 is given by the integral $W = \int_{x_1}^{x_2} \mathbf{F} \cdot d\mathbf{r}$. In this case, the force is $\mathbf{F} = (3x^2 - 2x)\mathbf{i}$ and the displacement is along the x-axis, so $d\mathbf{r} = dx\mathbf{i}$. The particle moves from $x_1 = 2$ m to $x_2 = 4$ m. Therefore, the work done is: $W = \int_2^4 (3x^2 - 2x)dx$ Integrating term by term: $W = \left[\frac{3x^3}{3} - \frac{2x^2}{2}\right]_2^4$.

26 [Foundational] • Target: Decomposing a vector into parallel and perpendicular parts

In the diagram below, lines AB and CD are parallel. Line EF is a transversal intersecting AB at G and CD at H. If the measure of angle AGH is $(2x + 10)^\circ$ and the measure of angle GHD is $(3x - 5)^\circ$, find the value of x.

a. 15

b. **25** ✓

c. 35

d. 45

Since lines AB and CD are parallel and EF is a transversal, angle AGH and angle GHD are consecutive interior angles. Consecutive interior angles are supplementary, meaning their measures add up to 180°. Therefore, we can set up the equation: $(2x + 10)^\circ + (3x - 5)^\circ = 180^\circ$ Combine like terms: $5x + 5 = 180$ Subtract 5 from both sides: $5x = 175$ Divide by 5: $x = \frac{175}{5}$ $x = 35$ Let's check the angles: Angle AGH = $2(35) + 10 = 70 + 10 = 80^\circ$ Angle GHD =.

27 [Foundational] • Target: Deriving the dot product angle formula

In the figure below, ΔABC is a right triangle with right angle at B, and altitude $\overline{BD}$ is drawn perpendicular to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of $\overline{BD}$?

a. $\sqrt{13}$

b. **6** ✓

c. 6.5

d. 13

By the Geometric Mean Theorem (Altitude Rule) for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments of the hypotenuse: $BD^2 = AD \cdot DC$ Given $AD = 4$ and $DC = 9$: $BD^2 = 4 \cdot 9 = 36 \Rightarrow BD = \sqrt{36} = 6$

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28 [Foundational] • Target: The Cauchy-Schwarz inequality (preview)

Prove the product formula: $\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)]$.

a. Only if $A = B$

b. Use integration

c. Not true

d. **From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓ ✓**

Applying Trigonometric Functions principles, Standard product-to-sum. Therefore, the verified correct option is From $\cos(A-B) = \cos A\cos B + \sin A\sin B$ and $\cos(A+B) = \cos A\cos B - \sin A\sin B$. Add and halve. ✓.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Dot products with variable components

Prove the sum-to-product formula: $\sin A + \sin B = 2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$.

a. Not true

b. Only for $A = B$

c. **Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓ ✓**

d. Use calculus

Applying Trigonometric Functions principles, Standard derivation. Therefore, the verified correct option is Set $A = M+N, B = M-N$; then $\sin(M+N)+\sin(M-N) = 2\sin M\cos N$. Substitute $M = (A+B)/2, N = (A-B)/2$. ✓.

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30 [Intermediate] • Target: A capstone dot product application

State the sum-to-product formula for $\cos A - \cos B$.

a. $2\cos\frac{A+B}{2}\cos\frac{A-B}{2}$

b. $2\sin\frac{A+B}{2}\cos\frac{A-B}{2}$

c. $2\cos\frac{A+B}{2}\sin\frac{A-B}{2}$

d. $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ ✓

Applying Trigonometric Functions principles, Standard sum-to-product. Therefore, the verified correct option is $-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$.

31 [Intermediate] • Target: Computing dot products and angles

Using the product-to-sum formula, simplify $\sin A\cos B$.

a. $\cos(A-B)$

b. $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$ ✓

c. $\sin(A+B)$

d. $\frac{1}{2}[\cos(A-B)-\cos(A+B)]$

Applying Trigonometric Functions principles, Product-to-sum formula. Therefore, the verified correct option is $\frac{1}{2}[\sin(A+B)+\sin(A-B)]$.

32 [Intermediate] • Target: Projections and work

Angles S and T are supplementary. If $\angle S = 34^\circ$, find $\angle T$.

a. 139

b. 143

c. 145

d. **146** ✓

Supplementary angles sum to 180°: $\angle T = 180^\circ - 34^\circ = 146^\circ$.

33 [Intermediate] • Target: Properties of the dot product

Angles M and N are complementary. If $\angle M = 69^\circ$, find $\angle N$.

a. **21** ✓

b. 25

c. 16

d. 18

Complementary angles sum to 90°: $\angle N = 90^\circ - 69^\circ = 21^\circ$.

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34 [Intermediate] • Target: Angle classification

Compute the dot product $\mathbf{u} \cdot \mathbf{v}$ for $\mathbf{u} = \langle 2, 3 \rangle$ and $\mathbf{v} = \langle 2, 5 \rangle$.

a. **19** ✓

b. 22

c. 16

d. 14

$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 = (2)(2) + (3)(5) = 4 + 15 = 19$.

35 [Intermediate] • Target: Scalar projection and further applications

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. Two co-interior (same-side interior) angles measure $(4x + 4)^\circ$ and 120°. Solve for x.

a. 21

b. 8

c. 20

d. **14** ✓

Co-interior angles are supplementary: $(4x + 4) + 120 = 180 \Rightarrow x = 14$.

36 [Intermediate] • Target: Decomposing a vector into parallel and perpendicular parts

Two lines intersect, forming vertical angles. If one angle measures 114°, what is the measure of its vertical angle?

a. 116

b. 120

c. 119

d. **114** ✓

Vertical angles are congruent, so the vertical angle also measures 114°.

37 [Intermediate] • Target: Deriving the dot product angle formula

Given matrix $A = (2, -1; 3, 3)$, what is the scalar product $3A$?

a. $(5, 2; 6, 6)$

b. $(6, -1; 3, 9)$

c. $(6, -3; 9, 9)$ ✓

d. $(6, 3; 9, 10)$

Multiply each element by 3: $3(2) = 6$, $3(-1) = -3$, $3(3) = 9$, $3(3) = 9$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The Cauchy-Schwarz inequality (preview) • Cross-Lesson: 2.02 Radian Measure and Arc Length

Lines $\ell_1 \parallel \ell_2$ are cut by a transversal. One angle measures $(3x + 5)^\circ$ and its corresponding angle measures 80°. Solve for x.

a. 33

b. 20

c. **25** ✓

d. 19

Corresponding angles are equal: $3x + 5 = 80 \Rightarrow x = 25$.

39 [Challenging] • Target: Dot products with variable components • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Evaluate $\int_0^1 xe^x\,dx$ using integration by parts.

a. -1

b. **1** ✓

c. $1 + e^2$

d. 2

$\int_0^1 xe^x\,dx = 1$.

40 [Challenging] • Target: A capstone dot product application • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

In ΔABC shown below, side AB has length 6, side BC has length 10, and the area of the triangle is $15\sqrt{3}$. If $\angle B$ is an obtuse angle, what is the length of side AC, denoted by x?

a. 14 ✓

b. $2\sqrt{19}$

c. $2\sqrt{34}$

d. 16

We solve the problem in two steps: 1. **Find the measure of $\angle B$:** The area of a triangle given two sides and the included angle is: Area $= \frac{1}{2} \cdot AB \cdot BC \cdot \sin B$ Substitute the known values: $15\sqrt{3} = \frac{1}{2} \cdot 6 \cdot 10 \cdot \sin B = 30 \sin B$ $\sin B = \frac{15\sqrt{3}}{30} = \frac{\sqrt{3}}{2}$ Since $\angle B$ is obtuse ($90^\circ < \angle B < 180^\circ$), we have: $\angle B = 180^\circ - 60^\circ = 120^\circ$ 2. **Apply the Law of Cosines to find $x = AC$:**.

41 [Challenging] • Target: Computing dot products and angles • Cross-Lesson: 3.01 The Unit Circle and Exact Values

The sum of interior angles of a hexagon is:

a. 540°

b. **720°** ✓

c. 900°

d. 360°

Applying Lines, Triangles & Polygons principles, $(6 - 2) \times 180 = 720°$ Therefore, the verified correct option is 720°.

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42 [Challenging] • Target: Projections and work • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A sector has radius 5 cm and angle 0.8 rad. Find the perimeter of the sector.

a. **14** ✓

b. 9

c. 4

d. 15

$P = 2r + r\theta = 2(5) + (5)(0.8) = 14$ cm

43 [Challenging] • Target: Properties of the dot product • Cross-Lesson: 3.03 Sum and Difference Formulas

In the figure below, ΔABC is a right triangle with hypotenuse $\overline{AB}$, and altitude $\overline{CD}$ of length h is drawn to $\overline{AB}$. The altitude divides the hypotenuse into two segments of lengths $AD = 9$ and $DB = 16$. A circle with center O_1 is inscribed in ΔACD, and a circle with center O_2 is inscribed in ΔBCD. What is the distance between the centers O_1 and O_2?

a. 5

b. $5\sqrt{2}$ ✓

c. 7

d. $7\sqrt{2}$

We solve the problem in five systematic steps: 1. **Determine altitude CD**: By the Geometric Mean Theorem for right triangles, the altitude to the hypotenuse satisfies:
$CD^2 = AD \cdot DB = 9 \cdot 16 = 144 \Rightarrow CD = 12.$ 2. **Find the side lengths of ΔACD and ΔBCD**: - In right triangle ΔACD, the legs are $AD = 9$ and $CD = 12$. By the Pythagorean theorem:
$AC = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225} = 15.$ - In right triangle ΔBCD, the legs are $DB = 16$ an.

44 [Challenging] • Target: Angle classification • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

In ΔABC, the side lengths opposite angles $A, B,$ and C are $a, b,$ and c, respectively. Given that $a = 7$, $b = 8$, and $\sin(B - A) = \frac{3}{5}\sin C,$ what is the length of side c?

a. 5 ✓

b. $3\sqrt{5}$

c. $\frac{5}{3}$

d. $2\sqrt{7}$

We synthesize angle sum/difference identities with the Law of Sines and the Law of Cosines. **Step 1: Relate $\sin C$ to $A + B$** Since the interior angles of a triangle sum to 180°, $C = 180^\circ - (A + B)$. Therefore: $\sin C = \sin(180^\circ - (A + B)) = \sin(A + B)$ Substituting this into the given equation gives: $\sin(B - A) = \frac{3}{5}\sin(A + B)$ **Step 2: Expand using compound angle formulas** Using the sine addition and subtraction identities.

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45 [Challenging] • Target: Scalar projection and further applications • Cross-Lesson: 4.01 Circles and Parabolas

A fair six-sided die is rolled twice. Let X be the random variable representing the sum of the outcomes of the two rolls. The probability distribution of X is given in the table below, where p is an unknown probability. $Sum\ (x), 2, 3, 4, 5, 6, 7; P(X = x), 136, p, 336, 436, 536, 636;$ What is the value of p?

a. $\frac{2}{36}$ ✓

b. $\frac{5}{36}$

c. $\frac{4}{36}$

d. $\frac{3}{36}$

The sum of all probabilities in a probability distribution must equal 1. We are given the probabilities for several sums, and we need to find the unknown probability p for the sum of 3. The sum of the given probabilities is: $P(X = 2) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) = \frac{1}{36} + \frac{3}{36} + \frac{4}{36} + \frac{5}{36} + \frac{6}{36} = \frac{1+3+4+5+6}{36} = \frac{19}{36}$ Since the total probability must be 1, we have: $P(X = 2) + P(X = 3) + P(X = 4) + P(X = 5) + P(X = 6) + P(X = 7) = 1$ Substituting the known value.

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